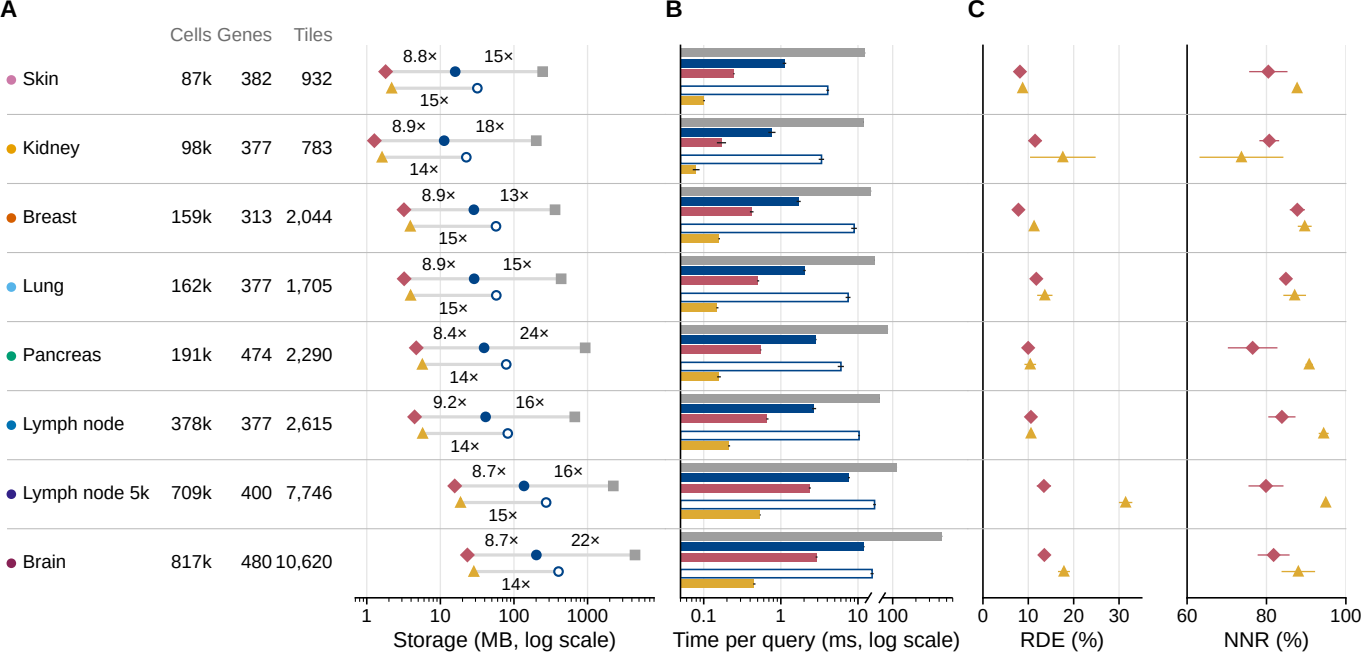
A
B
C
Cells
Genes
Tiles
Skin
87k
382
932
Kidney
98k
377
783
Breast
159k
313
2,044
Lung
162k
377
1,705
Pancreas
191k
474
2,290
Lymph node
378k
377
2,615
Lymph node 5k
709k
400
7,746
Brain
817k
480
10,620
8.8×
15×
15×
8.9×
18×
14×
8.9×
13×
15×
8.9×
15×
15×
8.4×
24×
14×
9.2×
16×
14×
8.7×
16×
15×
8.7×
22×
14×
1
10
100
1000
Storage (MB, log scale)
0.1
1
10
100
Time per query (ms, log scale)
0
10
20
30
RDE (%)
60
80
100
NNR (%)
Whole covariances
Spindle-naive
Spindle-naive (gene sets)
Spindle (whole tiles)
Spindle (gene sets)
Whole-matrix scan